

B.Tech. Examination-2023-24
Electronics and Communication Engineering Department
 (Odd Semester Regular and Supplementary)
Course Title - Basic Electronics Engineering Course Code - ECEUGES01

Full Marks : 80

Time-3 hrs

(Answer any five, Symbols have their usual meaning)

1. (a) Explain electron-hole pair generation in intrinsic Si semiconductor at room temperature. 2
- ✓(b) Compare between metal, semiconductor and insulator with respect to their band diagram. 4
- (c) How p-type semiconductor is formed? 3
- (d) Draw and explain the majority and minority carrier concentration curve of it. 3
- (e) Explain why in p-type semiconductor the Fermi level energy E_F is close to the valence band edge E_v of the band diagram. 4
2. (a) Establish the following relation between drift velocity, mobility and electric field for an electron in a semiconductor under thermal equilibrium. 8

$$v = \mu_n E$$
- (b) Write down the drift and diffusion current expression of electron in a semiconductor. 3
- (c) What is Einstein relation? 3
- (d) What is its value at room temperature? 2
3. (a) Write the Fermi-Dirac distribution function. 9
- ✓(b) Define Fermi-level. 2
- ✓(c) Show that Fermi function is symmetrical about E_F i.e. $f(E_F + \Delta E) = 1 - f(E_F - \Delta E)$ 6
- (d) Write down the density of states function with energy E in 3D semiconductor crystal. 3
4. (a) Show that equilibrium concentrations of electrons (n_0) and holes (p_0) in a semiconductor can be expressed as :- 10

$$(i) n_0 = N_c e^{-\left(\frac{E_c - E_F}{KT}\right)} \quad (ii) p_0 = N_v e^{-\left(\frac{E_F - E_v}{KT}\right)}$$
- ✓(b) Show that the product of electron and hole concentrations under equilibrium is constant and can be expressed as :- $n_0 p_0 = n_i^2$. 3
- ✓(c) Draw and explain the ideal diode characteristics in forward bias and reverse bias. 3
5. (a) Write down Shockley diode equation for p-n junction diode. 2
- (b) Show that diode dynamic resistance in forward bias is $r_d = \frac{26mV}{I_D}$. 3
- (c) Show that cut-in voltage of p-n junction diode is expressed as : 3

$$V_Y = V - \eta V_T \left(1 - e^{-\frac{V}{\eta V_T}}\right)$$
- (d) Determine the diode dynamic resistance ' r_d ' and cut-in voltage ' V_Y ' of the forward-bias diode at room temperature with the following parameters :- 4+4
 $I_S = 1 \text{ nA}, \eta = 1$, near the region where the diode current is 2 mA .
6. ✓(a) Explain the formation of depletion region and barrier potential V_0 in a p-n junction diode under equilibrium. 8
- ✓(b) Explain with diagram the forward bias and reverse bias characteristics of p-n junction diode. 8
7. (a) Draw and explain the output waveform of the circuit in Fig. 1. 5
- (b) Draw and explain the output waveform of the circuit in Fig. 2. 6
- (c) What will be the output waveform of the circuit shown in Fig. 3. 5

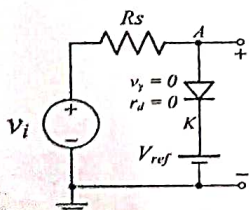


Fig. 1

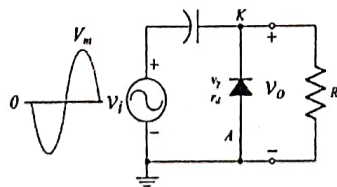


Fig. 2

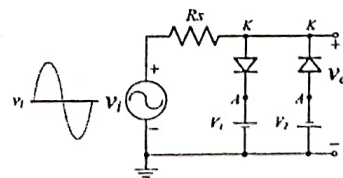


Fig. 3

8. (a) Explain the operation of full wave rectifier circuit using centre-tapped transformer. 7
- (b) Show that the output DC voltage (V_{dc}) and RMS value of full wave rectifier output is: (i) $V_{dc} = \frac{2V_m}{\pi}$ (ii) $V_{rms} = \frac{V_m}{\sqrt{2}}$ 6
- (c) What are the advantages of bridge rectifier over centre-tapped transformer full-wave rectifier? 3
- ✓ 9. (a) Explain different current components in a p-n-p transistor in active region with a suitable diagram. 6
- (b) Establish the relation between current amplification factors α and β . 4 •
- (c) Briefly explain the CB, CE and CC mode of operation in a transistor. 6
- ✓ 10. (a) What are the ideal characteristics of OPAMP? 4 •
- (b) Define CMRR and find the expression of CMRR in terms of input voltages. 5
- (c) Derive an expression of voltage gain of an inverting amplifier. 7
11. (a) Derive an expression for closed loop gain of a feedback amplifier. 6
- (b) Show that negative feedback in amplifier increases stability of the circuit. 5
- (c) Derive the expression of input impedance of voltage series feedback amplifier. 5
- ✓ 12. (a) Explain the construction and operation of n-channel JFET with suitable diagram. 8
- (b) Define ac drain resistance, transconductance and amplification factor of FET and find relation between them. 5 •
- ✓ (c) For an n-channel JFET $I_{DSS} = 8.7\text{mA}$, $V_p = -3\text{V}$ and $V_{GS} = -1\text{V}$. Find the value of I_D , g_{mo} and g_m . 3 •